Mini Project -2

Writeup

Project Title : A low cost mosquito trapping system using Arduino

Team Members :

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Abstract:

Mosquitoes transmit various diseases, including malaria, dengue, and Zika virus, posing significant public health risks. This project introduces an Arduino-based mosquito trapping system that uses frequencies generated by speaker to generate sound frequencies in the range of 200 to 500 Hz, which are believed to attract mosquitoes. Controlled by an Arduino microcontroller, a speaker that emits frequencies appealing to mosquitoes. The trapping mechanism incorporates sticky pads, suction unit and UV lights for efficient capture. By providing an eco-friendly solution that reduces reliance on chemical repellents, this project aims to mitigate mosquito populations and address the growing threat of mosquito-borne diseases.

Tools Required:

Arduino UNO, speaker, NPN transistor 2N3904 or 2N2222, 1N4001 diode, 9v dc motor, L293 motor driver, sticky pads, 5mm violet led, UV light

Method to be used:

The stimuli for attracting female mosquitoes are lights, body, warm temperature, CO_2 and particular frequencies. The mosquitoes get attracted when frequency is generated in the range of (200Hz to 500 Hz) using speaker. Due to this frequency, the mosquito come near the speaker, where the sticky pads are placed and the suction unit, suck the mosquito and put into the box. In addition to sound frequency , it will be more effective when we add other attractive stimuli like UV lights ,as they have higher wavelength these are more effective for attracting mosquito.

Beneficiary:

- Cost effective
- Low maintenance
- Health protection
- Targeted attraction
- Eco-Friendly solution
- Reduced chemical exposure
- User-Friendly technology

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Block Diagram:

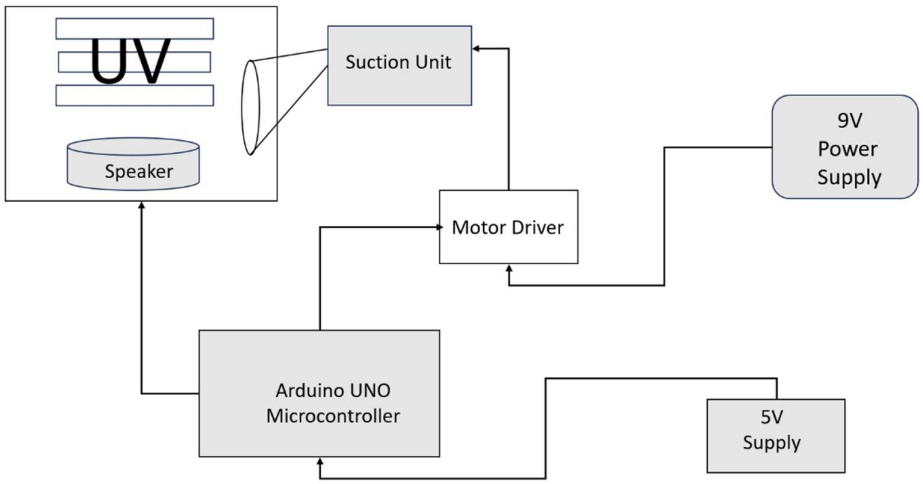

Fig. Arduino Based Mosquito Trapping System using Sound Frequency

Timeline:

Content	January				February				March				April	
	W-1	W-2	W-3	W-4	W-1	W-2	W-3	W-4	W-1	W-2	W-3	W-4	W-1	W-2
Literacy survey														
Purchase of components														
Hardware connection														
Result and execution														
Report submission														

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